

Dominik Hollidt

PhD Candidate in Computer Science at ETH



EDUCATION

PhD Candidate / December 2023 - now

SIPLAB - Zürich, SUI

Working on Machine Learning for egocentric human Motion Estimation and Understanding.

Master Thesis / March 2023 - August 2023

MIT (CSAIL, Medical Vision Group) - Cambridge, USA

Worked on deep 3D semantic segmentation via field-to-field transformation and neural radiance fields using 2D annotation only.

Master Studies / September 2020 - November 2023

ETH - Zürich, CH

Studying at one of the top universities for computer science with the major in machine intelligence and minor in theoretical computer science.

Machine Intelligence

Theoretical Computer Science

Exchange Student / August 2019 - June 2020

KTH Royal Institute of Technology - Stockholm, SE

Completed 90 ECTS with straight A's.

Wrote Bachelor Thesis about *Optical Flow Algorithms for Event Based Cameras* graded with 1.3 German Score. Created real-world optical flow data set for event-based cameras to investigate the impact of resolution on different optical flow algorithms.

Machine Learning

Computer Vision

Robotics

AI

Bachelor Student / Sept. 2017 - June 2020

Technical University Berlin - Berlin, GER

Graduated in top 3% with 1.3 German score.



PROJECTS

- **ETH Formula Student AMZ (09.2020 - 09.2022):** Member of autonomous racing car team. Developed LiDAR-based cone detection, multi-LiDAR calibration, and CI/CD infrastructure. Built a photorealistic simulator using Unreal Engine for end-to-end autonomous driving validation. Business Team Lead. 3rd place at FS Germany Driverless.
- **KTH Formula Student (08.2019 - 06.2020):** Perception team of autonomous racing car team. Implemented fast point cloud clustering for cone detection.

Email dominik.hollidt@googlemail.com

Linkedin [dominik-hollidt](#)

Website dominikvincent.github.io/



PUBLICATIONS

- **D. Hollidt**, T. Bendinelli, & C. Holz. *Ultra Diffusion Poser: Diffusion-Based Human Motion Tracking from Sparse Inertial Sensors and Ranging-based Between-sensor Distances*. CVPR 2026.
- Y. Xue, J. Jiang, R. Armani, **D. Hollidt**, Y.-C. Liao, & C. Holz. *Group Inertial Poser: Multi-Person Pose and Global Translation from Sparse Inertial Sensors and Ultra-Wideband Ranging*. ICCV 2025.
- **D. Hollidt**, P. Streli, J. Jiang, Y. Haghighi, C. Qian, X. Liu, & C. Holz. *EgoSim: An Egocentric Multi-View Simulator and Real Dataset for Body-Worn Cameras during Motion and Activity*. NeurIPS 2024.
- **D. Hollidt**, C. Wang, P. Golland, & M. Pollefeys. *Geometry Aware Field-to-field Transformations for 3D Semantic Segmentation*. 2023.



INTERSHIPS

ETH SIP Lab (12.2023-now):

- Research Internship in Human-Computer Interaction
- Working on Gesture detection via IMU and VPUs

Daedalean (06.2022-10.2022):

- Generation of synthetic flight scenarios from historical flight data for intruder detection
- Statistical analysis of historical flight track data
- Color grading for synthetic flight scenarios to reduce the sim-2-real gap
- Work on data evaluation pipeline and tooling
- Implementation of upload station for automated upload of test data to the cloud.

AVM (06.2020-09.2020):

- Performed data analysis of customer support data and automatically linked it to crashes and bugs
- Automated data summaries via Spark and Zeppelin and performed WiFi speed tests via statistical tests
- Provided a tool to see feature relevance via decision trees to allow for better understanding of bug-causing configurations.



Awards and Honors

- Spickenreuther Foundation Scholarship for Master's Thesis at MIT
- Swiss-European Mobility Programme for Master's Thesis at MIT
- VDI Honoring: For excellent graduation
- ERASMUS Scholarship
- IT-Talents Scholarship
- Nominated for "Stipendium des deutschen Volkes" due to excellent academic performance